Date & Time Functions in Excel

Complete Notes for Data Analytics | Interview-Oriented | Business Implementation

Date and time functions are essential in Data Analytics because almost every real business dataset contains time-based information such as:

- Order dates
- Employee joining dates
- Invoice dates
- Delivery deadlines
- Project start and completion dates
- Login and logout times
- Service ticket timestamps
- Payment due dates

A Data Analyst uses date and time functions to answer questions such as:
How many days did it take to close a ticket?
Is the ticket overdue?
How long has an employee worked for the company?
Which month generated the most tickets?
How many working days were required to complete a task?

Dataset Used in This Lecture

The dataset contains the following columns:

Column	Description
Ticket ID	Unique identifier for each service ticket
Employee Name	Employee responsible for handling the ticket
Department	Department of the employee
Start Date	Date on which work started
Start Time	Time at which work started
End Date	Date on which work was completed
End Time	Time at which work was completed
Due Date	Expected completion deadline
Joining Date	Employee's joining date
Region	Business region

Assuming the columns are arranged as follows:

Column	Field
A	Ticket ID
B	Employee Name
C	Department
D	Start Date
E	Start Time
F	End Date
G	End Time
H	Due Date

I	Joining Date
J	Region

Classification of Date & Time Functions

For this lecture, I recommend organizing the functions into these categories:

Category	Functions
Current Date & Time	TODAY, NOW
Date Creation	DATE, DATEVALUE
Date Extraction	DAY, MONTH, YEAR
Time Creation & Extraction	TIME, TIMEVALUE, HOUR, MINUTE, SECOND
Date Difference	DAYS, DATEDIF
Working Day Analysis	NETWORKDAYS, WORKDAY
Month-Based Calculations	EDATE, EOMONTH
Week Analysis	WEEKDAY, WEEKNUM
Date & Time Formatting	TEXT

We'll start with the first four foundational functions.

Module 1: Current Date & Time Functions

1. TODAY Function

The TODAY function returns the **current system date**.
It does not require any arguments.

Purpose

It is used when a calculation needs to automatically update based on the current date.

Syntax

=TODAY()
There are **no arguments inside the parentheses**.

Example

Suppose today's date is:
10-Jul-2026
Formula:
=TODAY()
Result:
10-Jul-2026
The result automatically changes when the workbook recalculates on another day.

Business Use Case: Employee Tenure

Suppose the employee's Joining Date is in I2.
Formula:
=TODAY()-I2
This returns the total number of days since the employee joined the company.

Business Use Case: Check Whether a Ticket Is Overdue

Suppose the Due Date is in H2.
Formula:
=IF(TODAY()>H2,"Overdue","Not Overdue")

Logic Explained

- TODAY() returns the current date.
- H2 contains the ticket's due date.
- TODAY()>H2 checks whether the current date has passed the deadline.
- If TRUE → "Overdue"
- If FALSE → "Not Overdue"

Important Points

- TODAY() does not take any arguments.
- It returns only the current date, not the current time.
- It is a dynamic or volatile date function.
- The result updates when Excel recalculates the workbook.

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2. NOW Function

The NOW function returns the **current system date and current system time**.

Purpose

It is used when both the current date and exact time are required.

Syntax

=NOW()
No arguments are required.

Example

Formula:
=NOW()
Possible result:
10-Jul-2026 02:45:30 PM

Business Use Cases

NOW is useful for:

- Report generation timestamps
- Ticket monitoring
- SLA tracking
- Login/logout analysis
- Dashboard refresh timestamps
- Real-time operational reports

Business Example: Report Timestamp

="Report Generated: "&TEXT(NOW(),"dd-mmm-yyyy hh:mm AM/PM")

Possible result:
Report Generated: 10-Jul-2026 02:45 PM

Important Points

- NOW() returns both date and time.
- It does not require arguments.
- It updates when the workbook recalculates.
- It does not continuously update every second unless Excel recalculates.

Difference Between TODAY and NOW

Feature	TODAY	NOW
Returns current date	Yes	Yes
Returns current time	No	Yes
Requires arguments	No	No
Dynamic	Yes	Yes
Example	10-Jul-2026	10-Jul-2026 02:45 PM

Module 2: Date Creation & Conversion Functions

3. DATE Function

The DATE function creates a valid Excel date using separate **year, month, and day values**.

Purpose

It is used when year, month, and day are stored separately or when a date must be dynamically constructed.

Syntax

=DATE(year,month,day)

Syntax Explanation

Argument	Meaning

year	Year of the date
month	Month number from 1 to 12
day	Day of the month

Example

=DATE(2026,7,10)
Result:
10-Jul-2026

Business Example

Suppose:

A	B	C
Year	Month	Day
2026	7	10

Formula:
=DATE(A2,B2,C2)
Result:
10-Jul-2026

Why Use DATE Instead of Typing the Date Manually?

Because DATE ensures Excel creates a genuine date value that can be used in:

- Date calculations
- Sorting
- Filtering
- Pivot Tables
- Dashboards
- Date comparisons

Advanced Behaviour

Excel can automatically adjust overflowing month values.
For example:
=DATE(2026,13,1)
Result:
01-Jan-2027
Because month 13 means one month after December 2026.

Important Points

- DATE returns an actual Excel date.
- All three arguments are required.
- It is useful when date components exist in separate columns.
- It avoids ambiguity caused by regional date formats.

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4. DATEVALUE Function

The DATEVALUE function converts a **date stored as text** into an Excel date serial number.

Purpose

It is particularly useful when dates imported from CSV files, ERP systems, web sources, or external databases are recognized as text instead of valid Excel dates.

Syntax

=DATEVALUE(date_text)

Syntax Explanation

Argument	Meaning
date_text	A text string representing a valid date

Example

Suppose cell A2 contains:
10-Jul-2026
But Excel recognizes it as text.
Formula:
=DATEVALUE(A2)
Excel may internally return:
46213
After applying a date format, it displays as:
10-Jul-2026

Why Does Excel Return a Number?

Excel internally stores dates as serial numbers. The displayed date is simply formatting applied to that underlying numeric value.

Business Scenario

An ERP export contains:
"15-Mar-2026"
Because it is stored as text, some date calculations may fail.
Formula:
=DATEVALUE(A2)
The text is converted into a genuine Excel date value.

Common Uses

- CSV imports
- ERP exports
- CRM data
- Web-scraped datasets
- External database exports
- Data cleaning before analysis

Important Limitation

DATEVALUE can depend on the computer's regional date settings. Ambiguous dates such as:
05/06/2026
could mean either:

- 5 June 2026, or
- May 6, 2026

Therefore, analysts should avoid ambiguous date formats whenever possible.

DATE vs DATEVALUE

DATE	DATEVALUE
Creates a date from separate components	Converts text into a date
Takes year, month, and day	Takes date text
=DATE(2026,7,10)	=DATEVALUE("10-Jul-2026")
Useful for constructing dates	Useful for cleaning imported data

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Module 1 & 2 Summary

Function	Purpose	Syntax
TODAY	Returns current date	=TODAY()
NOW	Returns current date and time	=NOW()
DATE	Creates a date from year, month, and day	=DATE(year,month,day)
DATEVALUE	Converts text representing a date into an Excel date value	=DATEVALUE(date_text)

Module 3: Date Extraction & Week Analysis Functions

In the previous modules, we covered:

- TODAY
- NOW
- DATE
- DATEVALUE

Now we'll move to functions used to **extract specific components from dates and perform week-based analysis**.

These functions are highly useful in Data Analytics for:

- Year-wise sales analysis
- Monthly performance reports
- Day-level trend analysis
- Identifying weekdays and weekends
- Weekly KPI reporting
- Creating Pivot Tables and dashboards
- Building calculated columns for Power BI-style analysis in Excel

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5. DAY Function

The DAY function extracts the **day of the month** from a valid Excel date.
It returns a number between **1 and 31**.

Purpose

Used when an analyst needs to identify or extract the day component from a complete date.

Syntax

=DAY(serial_number)

Syntax Explanation

Argument	Meaning
serial_number	A valid Excel date, cell reference containing a date, or date serial number

Example

Suppose D2 contains:
15-Mar-2026
Formula:
=DAY(D2)
Result:
15

Example Using Our Business Dataset

Suppose the Start Date in D2 is:
08-Jul-2026
Formula:
=DAY(D2)
Result:
8

Business Use Cases

The DAY function can be used to:

- Analyze transactions by day of the month.
- Identify month-end activities.
- Group orders into early-month and late-month periods.
- Check whether deadlines fall on specific days.
- Create custom reporting periods.

Example: Classify Tickets by Month Period

=IF(DAY(D2)<=10,"Early Month",IF(DAY(D2)<=20,"Mid Month","Late Month"))

Logic

- Days 1–10 → Early Month
- Days 11–20 → Mid Month
- Days 21–31 → Late Month

This is a practical example of combining DAY with IF.

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6. MONTH Function

The MONTH function extracts the **month number** from a valid Excel date.

It returns a number from:
1 to 12
Where:

Number	Month
1	January
2	February
3	March
...	...
12	December

Purpose

Used to identify the month associated with a particular date.

Syntax

=MONTH(serial_number)

Syntax Explanation

Argument	Meaning
serial_number	A valid Excel date or cell reference containing a date

Example

Suppose D2 contains:
15-Mar-2026
Formula:
=MONTH(D2)
Result:
3

Business Example Using the Dataset

Suppose:
Start Date = 25-Nov-2025
Formula:
=MONTH(D2)
Result:
11

Business Use Cases

The MONTH function is useful for:

- Monthly ticket analysis
- Monthly sales trends
- Revenue reporting
- Customer acquisition analysis
- Expense analysis
- Seasonal trend identification

Example: Identify Quarter from Month

=IF(MONTH(D2)<=3,"Q1",IF(MONTH(D2)<=6,"Q2",IF(MONTH(D2)<=9,"Q3","Q4")))

Result logic:

Month	Quarter
January–March	Q1
April–June	Q2
July–September	Q3
October–December	Q4

This is a useful interview-oriented business scenario combining MONTH with nested IF.

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7. YEAR Function

The YEAR function extracts the **year component** from a valid Excel date.

Purpose

Used to identify the year associated with a date for yearly analysis, comparison, and reporting.

Syntax

=YEAR(serial_number)

Syntax Explanation

Argument	Meaning
serial_number	A valid Excel date, cell reference, or date serial number

Example

Suppose D2 contains:
15-Mar-2026
Formula:
=YEAR(D2)
Result:
2026

Business Example

Suppose the Joining Date in I2 is:
18-Aug-2021
Formula:
=YEAR(I2)
Result:
2021

Business Use Cases

The YEAR function is commonly used for:

- Year-over-year analysis
- Employee joining-year analysis
- Financial-year reporting
- Customer acquisition trends

- Annual sales reports
- Cohort analysis

Example: Classify Employees by Joining Period

=IF(YEAR(I2)<2020,"Veteran Employee",IF(YEAR(I2)<=2023,"Experienced Employee","Recent Joiner"))
This creates three categories based on the employee's joining year.

Combining DAY, MONTH and YEAR

Suppose D2 contains:
25-Dec-2025
Then:

Formula	Result
=DAY(D2)	25
=MONTH(D2)	12
=YEAR(D2)	2025

These functions **extract date components**. They do not change the original date.

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8. WEEKDAY Function

The WEEKDAY function returns a number representing the **day of the week** for a given date.
For example, depending on the selected return type:

- Monday can be represented by 1
- Tuesday by 2
- Wednesday by 3
- And so on

Purpose

Used to determine which day of the week a particular date falls on.

Syntax

=WEEKDAY(serial_number,[return_type])

Syntax Explanation

Argument	Meaning
serial_number	The date to evaluate
[return_type]	Optional argument that determines how weekdays are numbered

Most Important Return Types

Return Type	Week Starts On	Numbering
1 or omitted	Sunday	Sunday = 1, Saturday = 7
2	Monday	Monday = 1, Sunday = 7

For business analytics, I recommend teaching **return type 2**, because Monday-to-Sunday numbering is generally more intuitive for business-week analysis.

Example

Suppose D2 contains:
10-Jul-2026
Formula:
=WEEKDAY(D2,2)
Since 10 July 2026 is a Friday, the result is:
5
Because:

Day	Number
Monday	1
Tuesday	2
Wednesday	3
Thursday	4
Friday	5
Saturday	6
Sunday	7

Business Example: Identify Weekday or Weekend

=IF(WEEKDAY(D2,2)<=5,"Weekday","Weekend")

Logic

- Monday to Friday return 1 to 5 → Weekday
- Saturday and Sunday return 6 or 7 → Weekend

This is one of the most practical uses of WEEKDAY.

Business Example: Identify Exact Day Name

You could use:
=TEXT(D2,"dddd")
Result:
Friday
Or the shorter version:
=TEXT(D2,"ddd")
Result:
Fri
Important Difference
WEEKDAY returns a **numeric representation** of the weekday, while TEXT can return the **actual weekday name as text**.

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9. WEEKNUM Function

The WEEKNUM function returns the **week number of the year** for a given date.
For example:
01-Jan-2026
belongs to the first week of the year, so the result would generally be:

1

Purpose

Used for weekly reporting and performance analysis.

Syntax

=WEEKNUM(serial_number,[return_type])

Syntax Explanation

Argument	Meaning
serial_number	Date for which the week number is required
[return_type]	Determines which day the week begins on

Common Example

=WEEKNUM(D2,2)
Here:

- D2 = Date
- 2 = Week starts on Monday

Business Use Cases

The WEEKNUM function is useful for:

- Weekly sales analysis
- Weekly service-ticket reporting
- Employee productivity tracking
- Operations reporting
- Weekly KPI dashboards
- Project tracking

Example

Suppose:
Start Date = 10-Jul-2026
Formula:
=WEEKNUM(D2,2)
Result:
28
So the ticket was opened during approximately **Week 28 of 2026**, under Excel's Monday-start week numbering system.

WEEKDAY vs WEEKNUM

Function	Purpose	Example Result
WEEKDAY	Identifies the day within a week	Friday → 5
WEEKNUM	Identifies the week within a year	Week 28 → 28

Important Interview Point: WEEKNUM vs ISO Week Number

This distinction matters in professional analytics.
WEEKNUM and ISO week numbering are **not always identical**, especially around the beginning and end of a year.
If a business specifically requires **ISO 8601 week numbering**, Excel provides:
=ISOWEEKNUM(D2)
For beginner-level Date & Time functions, WEEKNUM is enough. But mentioning ISOWEEKNUM is valuable because some international organizations use ISO week standards.

Module 3 Summary

Function	Purpose	Syntax
DAY	Extracts day of the month	=DAY(serial_number)
MONTH	Extracts month number	=MONTH(serial_number)
YEAR	Extracts year	=YEAR(serial_number)
WEEKDAY	Returns day-of-week number	=WEEKDAY(serial_number,[return_type])
WEEKNUM	Returns week number of the year	=WEEKNUM(serial_number,[return_type])

Important formulas from this section

=DAY(D2)
=MONTH(D2)
=YEAR(D2)
=WEEKDAY(D2,2)
=WEEKNUM(D2,2)

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